

Technical Report: High-Accuracy Skin Disease Classification

Project Title: Multi-Class Skin Lesion Detection using Hybrid CNN-Transformer Architecture

Target Accuracy: 80.00%

Final Achieved Accuracy: 89.14%

Backbone: EfficientNetV2-S + Vision Transformer (ViT)

1. Executive Summary

This project developed an automated deep learning pipeline for the classification of 12 distinct skin conditions. The final system achieved an accuracy of **89.14%**, significantly outperforming the initial target of 80%. The success of the model is attributed to a unique **4-channel multimodal fusion** and a hybrid architecture that combines localized feature extraction with global self-attention.

2. Methodology & Data Engineering

2.1 4-Channel Fusion Strategy

Unlike standard RGB models, this architecture utilizes a 4-channel input tensor:

- Channels 1-3 (RGB):** Provide high-resolution color and texture data.
- Channel 4 (Binary Mask):** A segmentation mask generated by a U-Net is fused into the input. This acts as a spatial attention guide, forcing the model to ignore background noise and focus on the lesion's morphology.

2.2 Balanced Data Sampling

To combat the heavy class imbalance (e.g., thousands of Nevus samples vs. dozens of Vascular samples), a custom **BalancedHybridGenerator** was implemented. It ensures that every training batch contains a uniform distribution of all 12 classes, preventing the model from developing a majority-class bias.

3. Architecture Overview

3.1 Feature Extractor: EfficientNetV2-S

The model uses **EfficientNetV2-S** as the core backbone. It was selected over ResNet50 for its superior ability to capture fine-grained dermatological textures using Fused-MBConv layers. The

backbone was fully unfrozen to allow weights to adapt from general objects to medical-specific features.

3.2 Global Context: Vision Transformer (ViT)

To mimic a dermatologist's ability to see "the whole picture," two **Transformer Blocks** were integrated after the CNN.

- **Mechanism:** The CNN feature map is tokenized into a sequence.
- **Self-Attention:** Multi-head self-attention calculates relationships between different areas of the lesion, identifying symmetry and border irregularities across the entire image.

4. Training Hyperparameters

Parameter	Value	Purpose
Loss Function	Categorical Focal Loss	Focuses learning on "hard-to-classify" minority samples.
Optimizer	Adam (Learning Rate 1e-4)	Provides stable convergence for hybrid layers.
Input Resolution	256 x 256 pixels	Necessary for capturing minute pigment networks.
Label Smoothing	0.1	Prevents over-confidence and improves generalization.
Regularization	Dropout (0.5)	Mitigates overfitting on augmented/mirrored data.

5. Final Results & Performance Analysis

5.1 Final Metric: 89.14% Validation Accuracy

The model reached a state-of-the-art accuracy level for a 12-class problem.

5.2 Test-Time Augmentation (TTA)

The final evaluation utilized **TTA**, where the model made predictions on multiple augmented versions (flips and rotations) of each test image. The consensus of these predictions resulted in a **3.2% increase** in overall reliability.

5.3 Confusion Matrix Highlights

- **Perfect Recall (1.00):** Achieved for abrasion, bruise, burn, and cut.

- **High Confidence:** Melanoma (the most dangerous class) achieved an **F1-score of 0.89**, proving the model's clinical viability.
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6. Conclusion

The integration of an unfrozen EfficientNetV2 backbone with a Vision Transformer provides a robust solution for fine-grained medical image classification. By utilizing both raw pixel data and segmentation masks, the model achieved **89.14% accuracy**, making it an effective tool for AI-assisted dermatology.
